

FORMER TACOMA TIDEFLATS LANDFILL

1620 Lincoln Avenue, Tacoma, WA 98421

Port Parcels 30, 32, 37, 38, 95, 96, 97, 102, 104 and 105

Site Details

WA Department of Ecology FSID: 1240

WA Department of Ecology CSID: 3645

WA Department of Ecology SHARP Ranking(s): Medium

EPA Site ID: WAD988511844 **VCP Project ID:** WA1834

Cleanup Contaminants: Metals including arsenic and lead; petroleum products including diesel-range and oil-range organics and benzene; polycyclic aromatic hydrocarbons; polychlorinated biphenyls; chlorinated solvents; PFAS at some locations; landfill gas (methane) generated by decomposing organic waste

Current Remediation Status: Awaiting Cleanup. Planning and delineation; approximate centre of footprint 47.2653 N, 122.4051 W; reported extent more than 200 acres; period of operation early 1940s to 1964

Site Graphic(s)

Fifteen figures document the interpreted landfill boundary, the Port parcels it crosses, five subsurface cross sections, the contamination-point record, screening against Model Toxics Control Act cleanup levels, the Gog-le-hi-te habitat areas, and the landfill gas monitoring network.

Figure 1. Historic Landfill Extent. The boundary drawn on every figure in this fact sheet is an interpretation, not a surveyed limit. It was assembled by compiling the available boundary interpretations, the investigation-point record, and the historical reference material into a single geodatabase and then reconciling them against one another. Two published interpretations disagree: the Tacoma-Pierce County Health Department's original 1993 closed-landfill study produced the first of them, and the Department's revised 2010 mapping differs from it, reflecting continuing uncertainty about the true extent of disposal. Ecology's cleanup site record describes an extent of more than 200 acres. The boundary shown here should be read as the best current interpretation of a footprint that will continue to be refined as additional subsurface data are collected.



Figure 1. Historic Landfill Extent. *Interpreted footprint of the former Lincoln Avenue Landfill on Port of Tacoma parcels.*

Figure 2. Port Parcels Overlapping the Landfill. Ten Port of Tacoma parcels intersect the interpreted footprint: 30, 32, 37, 38, 95, 96, 97, 102, 104 and 105. Ownership across the wider footprint is mixed, taking in private industrial operators, railroads, public agencies and the Puyallup Tribe, and the practical consequence is that no single cleanup decision governs the former landfill. Obligations depend on which cleanup site record applies to a given parcel, which programme it sits under, and whether a voluntary cleanup or formal order is driving the work. Parcel boundaries here are shown to identify where Port interest and interpreted landfill extent overlap; they are not a statement of liability.

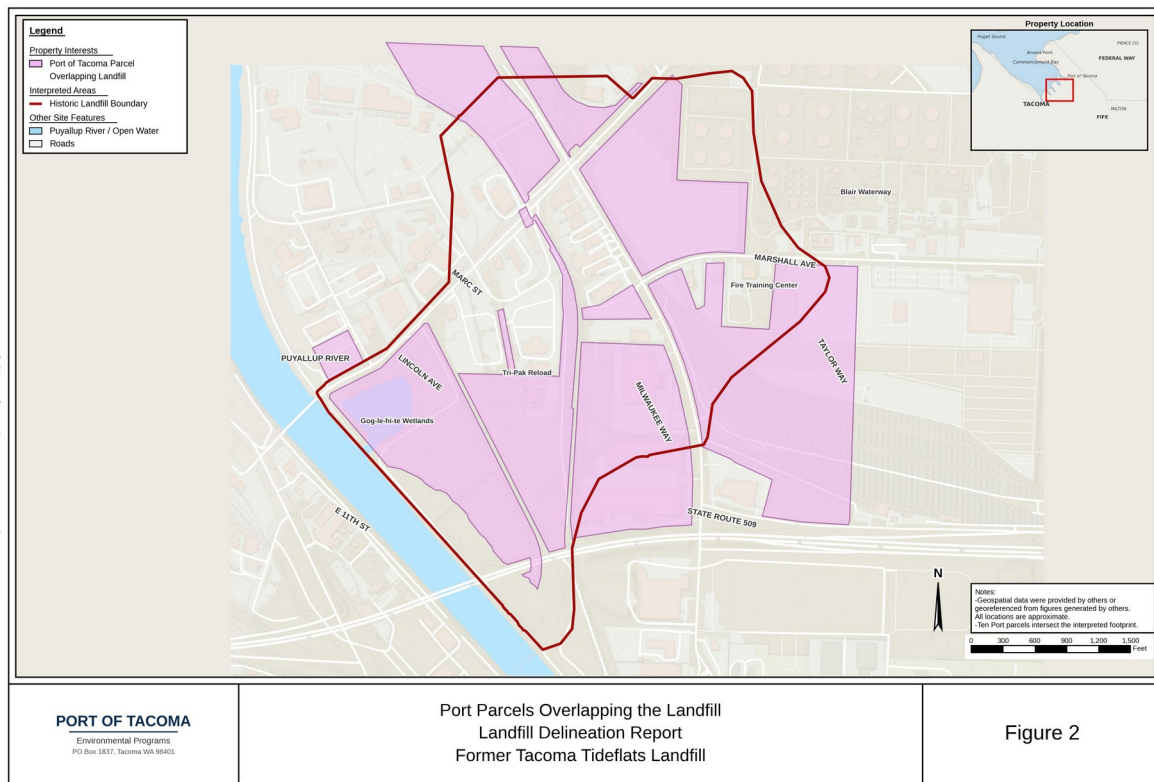


Figure 2. Port Parcels Overlapping the Landfill. Ten Port of Tacoma parcels within the interpreted landfill footprint.

Figure 3. Cross-Section Location Map. Five measured cross sections, A-A' through E-E', were constructed through the interpreted footprint using an automated ArcPy workflow. Each section line was drawn through a corridor of interest, and every investigation point falling within that corridor was projected onto the section line at its measured station. The corridor is 280 feet each side, so a section captures the locations near it rather than only those exactly on the line. The sections are measured rather than schematic: horizontal position comes from the projected station and depth from the reported investigation interval. Earlier schematic parcel sections were prepared during the same project and should not be confused with these.

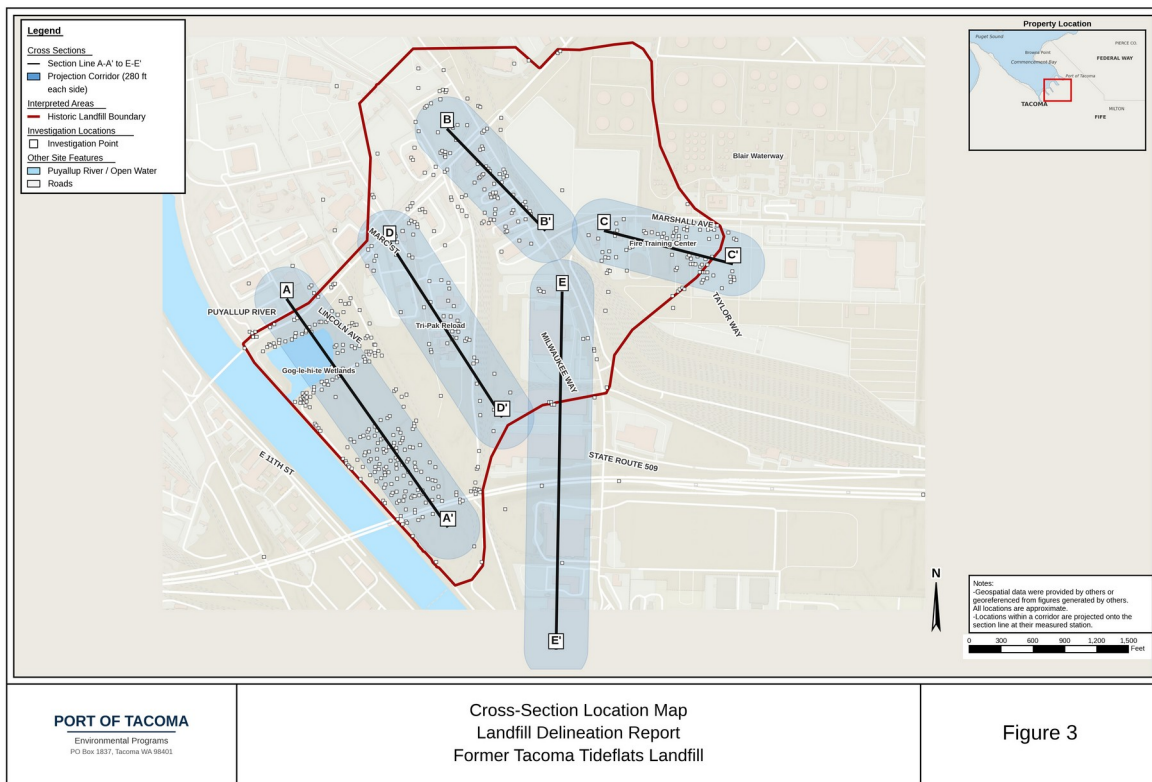
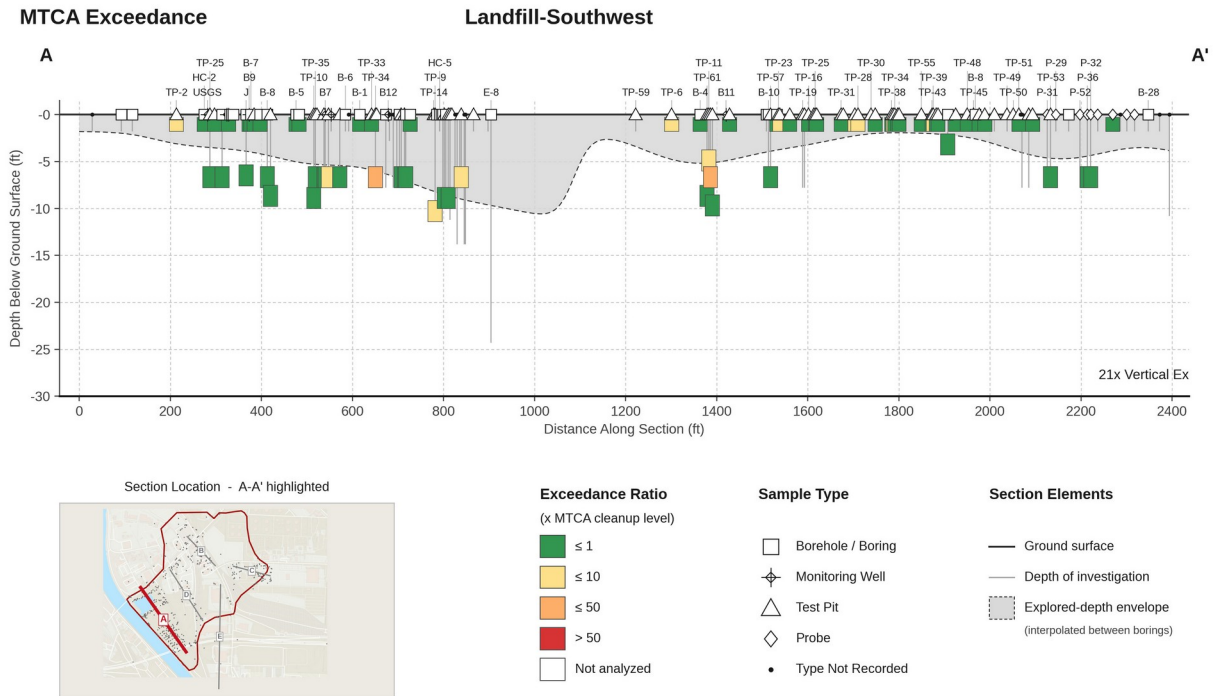


Figure 3. Cross-Section Location Map. Sections A-A' through E-E' and the 280-foot projection corridors used to build them.

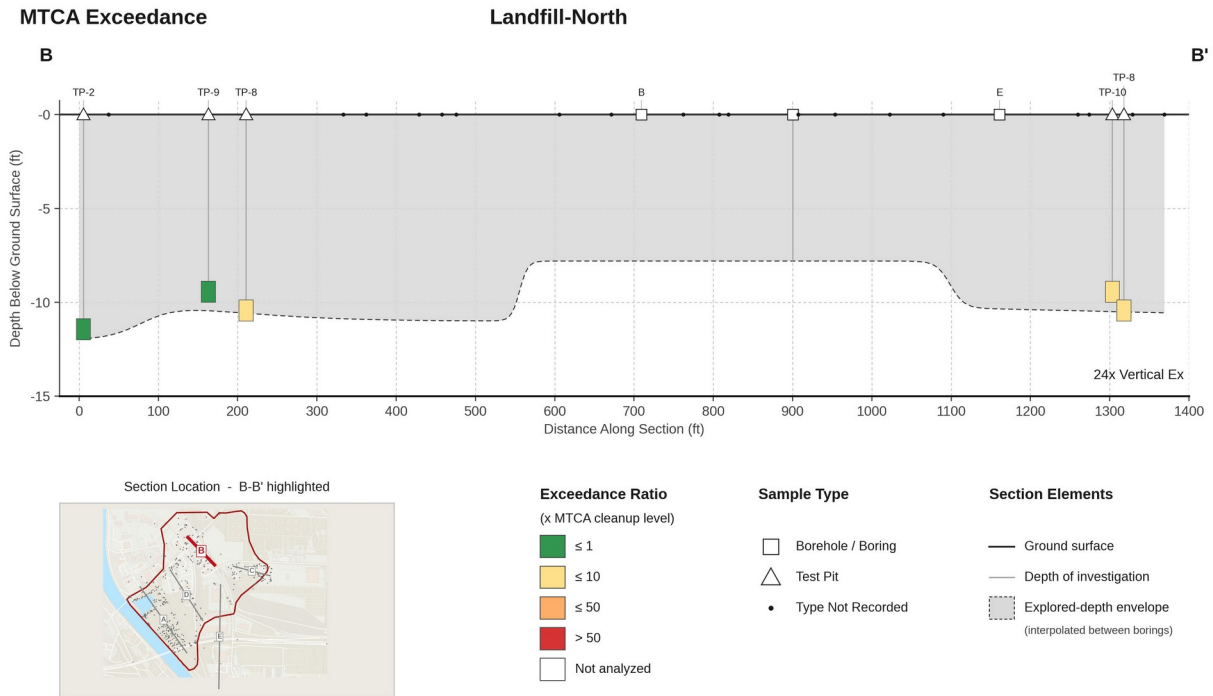
Figure 4. Cross Section A-A'. Section A-A' runs the Lincoln Avenue corridor in the southwest reach, from the riverbank near the Gog-le-hi-te Wetlands to the southeast, and is the best-populated section on the site with 135 locations over 2,394 feet. It also shows the clearest evidence of prior remediation: a large share of the interval blocks along it are recorded in the project GIS as Soil Removed and Monitored, reflecting the soil removal that accompanied habitat construction and the monitoring that followed. The dashed envelope is the explored depth, interpolated between borings, and is not a measured contact between fill and native ground.



Former Tacoma Tidelands Landfill Delineation Project | Figure 4 | 135 locations over 2,394 ft | 120 identified from the project contamination-point table | depths in ft below ground surface

Figure 4. Cross Section A-A'. Lincoln Avenue corridor, southwest reach. 135 locations over 2,394 feet.

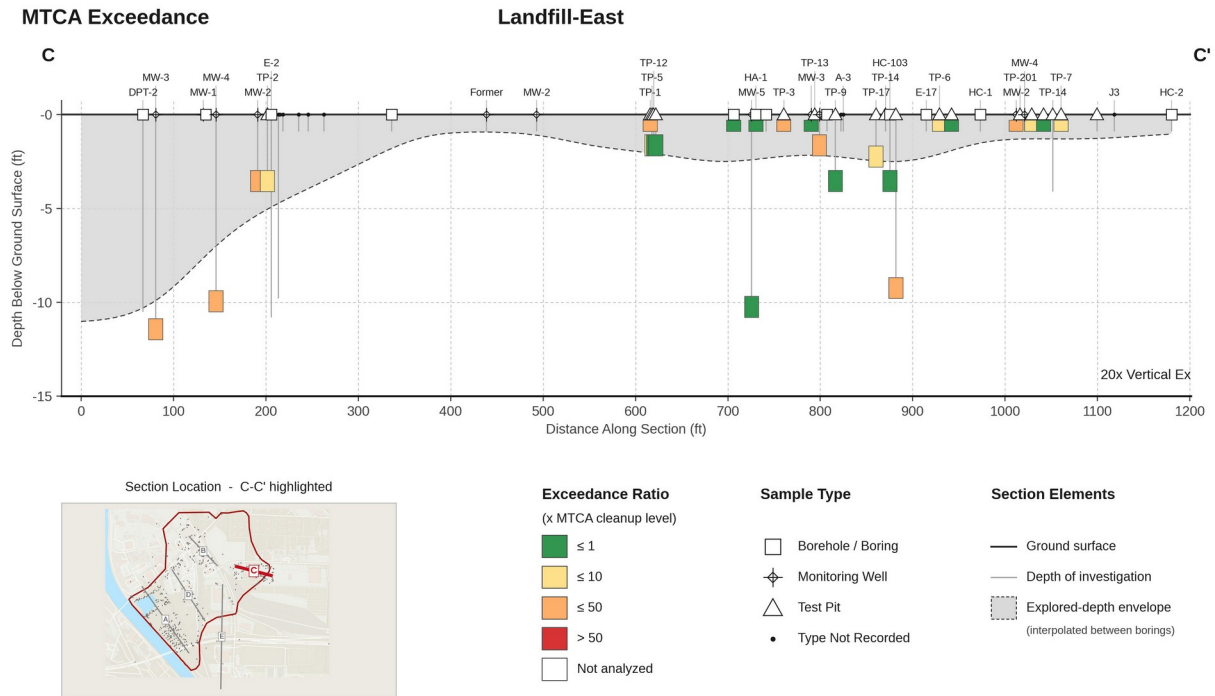
Figure 5. Cross Section B-B'. Section B-B' crosses the northern part of the footprint between Lincoln Avenue and Marc Street, 27 locations over 1,369 feet. Coverage here is thinner than on A-A' and the analytical record thinner still. Station identifiers on this section are recorded in the source plots by Ecology facility site ID rather than by report ID, so only a few can be labelled; the unlabelled locations are real investigations whose identifiers are not recoverable from the material supplied, which is a limitation of the source record rather than of the section.



Former Tacoma Tideflats Landfill Delineation Project | Figure 5 | 27 locations over 1,369 ft | 9 identified from the project contamination-point table | depths in ft below ground surface

Figure 5. Cross Section B-B'. Northern footprint between Lincoln Avenue and Marc Street. 27 locations over 1,369 feet.

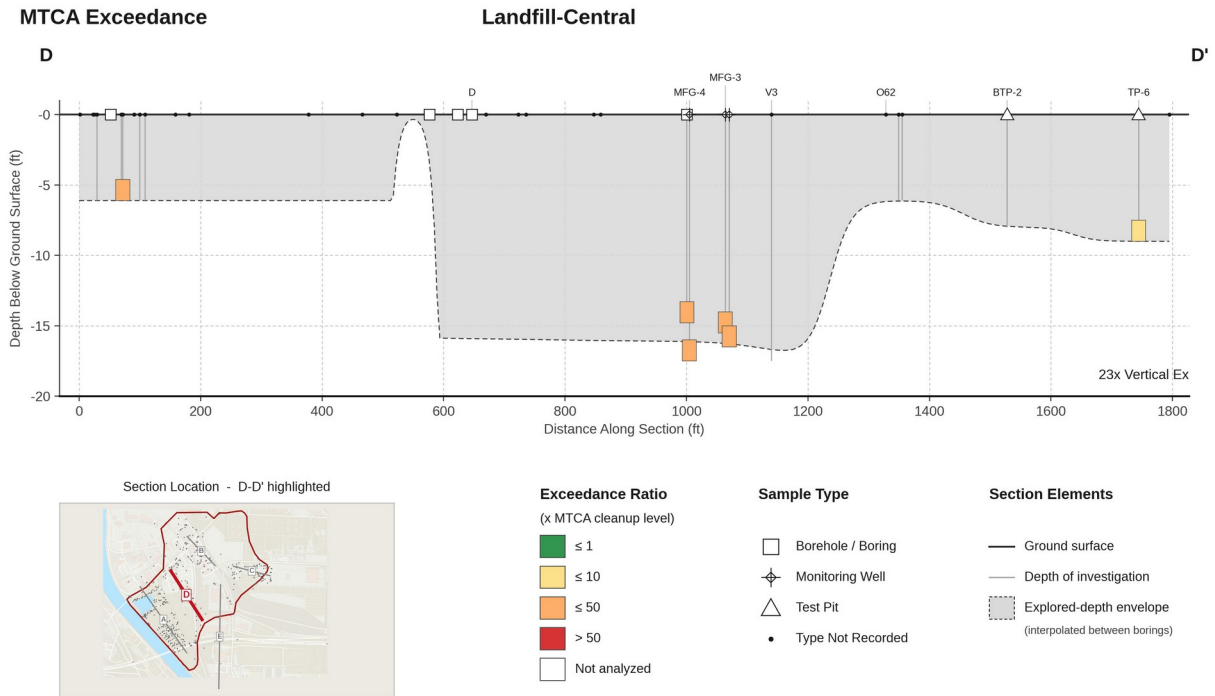
Figure 6. Cross Section C-C'. Section C-C' crosses the eastern footprint through the Fire Training Center reach, 51 locations over 1,180 feet. This is the densest run of confirmed exceedances on the site. Petroleum and polycyclic aromatic hydrocarbon results occur together at several of the same locations, and gasoline or diesel is identified as the fuel type at several of them, which is consistent with the petroleum handling and railroad-related uses that followed landfill closure rather than with municipal refuse alone.



Former Tacoma Tideflats Landfill Delineation Project | Figure 6 | 51 locations over 1,180 ft | 45 identified from the project contamination-point table | depths in ft below ground surface

Figure 6. Cross Section C-C'. Eastern footprint through the Fire Training Center reach. 51 locations over 1,180 feet.

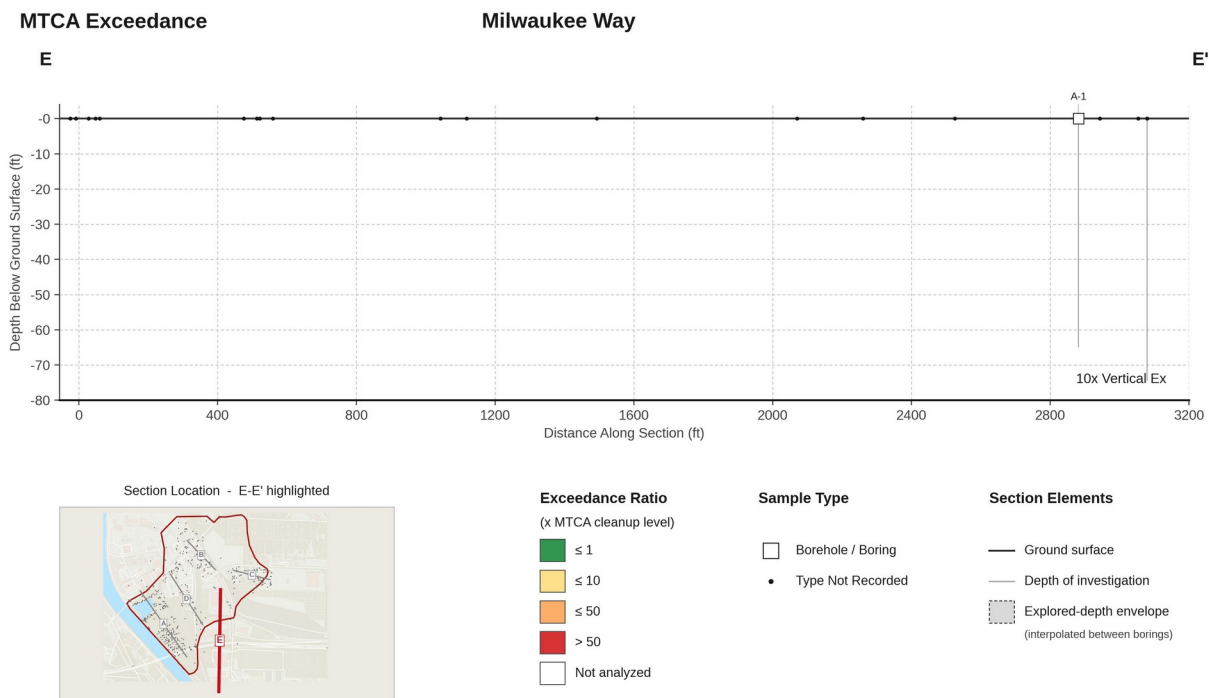
Figure 7. Cross Section D-D'. Section D-D' runs through the central footprint and the Tri-Pak Reload reach, 34 locations over 1,795 feet. The highest single concentration ratio recorded anywhere on the site sits in this part of the footprint: a polycyclic aromatic hydrocarbon result at 225 times the applicable cleanup level, isolated from the eastern cluster and paired with a petroleum result of only 1.5 times. An isolated high result of that kind is a strong argument for treating the surrounding area as under-characterised rather than as bounded.



Former Tacoma Tideflats Landfill Delineation Project | Figure 7 | 34 locations over 1,795 ft | 16 identified from the project contamination-point table | depths in ft below ground surface

Figure 7. Cross Section D-D'. Central footprint through the Tri-Pak Reload reach. 34 locations over 1,795 feet.

Figure 8. Cross Section E-E'. Section E-E' follows the Milwaukee Way corridor north to south, 19 locations over 3,079 feet, and is the longest and most sparsely sampled of the five. It also carries the deepest documented investigation on any section, reaching 74.5 feet. Because the spacing between locations is wide, the explored-depth envelope on this section is the least constrained of the set and should be read as indicative only.



Former Tacoma Tideflats Landfill Delineation Project | Figure 8 | 19 locations over 3,079 ft | 2 identified from the project contamination-point table | depths in ft below ground surface

Figure 8. Cross Section E-E'. Milwaukee Way corridor, north to south. 19 locations over 3,079 feet; deepest investigation 74.5 feet.

Figure 9. Contamination Points by Observation Category. The compiled inventory holds 559 mapped investigation locations, symbolised here in the nine observation categories used in the project GIS. Analytical coverage is markedly uneven. A substantial share of the record consists of historical boreholes, monitoring wells and test pits for which no laboratory results were preserved, and these are mapped as No Analytical Data. That category must not be read as evidence of clean soil: it records the absence of a preserved result, and it is the single largest class on the map. Equally, a single depth-specific result must not be extended through an entire boring. The four interpretive classes used through this fact sheet -- confirmed exceedance, detected below cleanup level, landfill material observed, and no analytical data -- are deliberately kept separate for this reason.

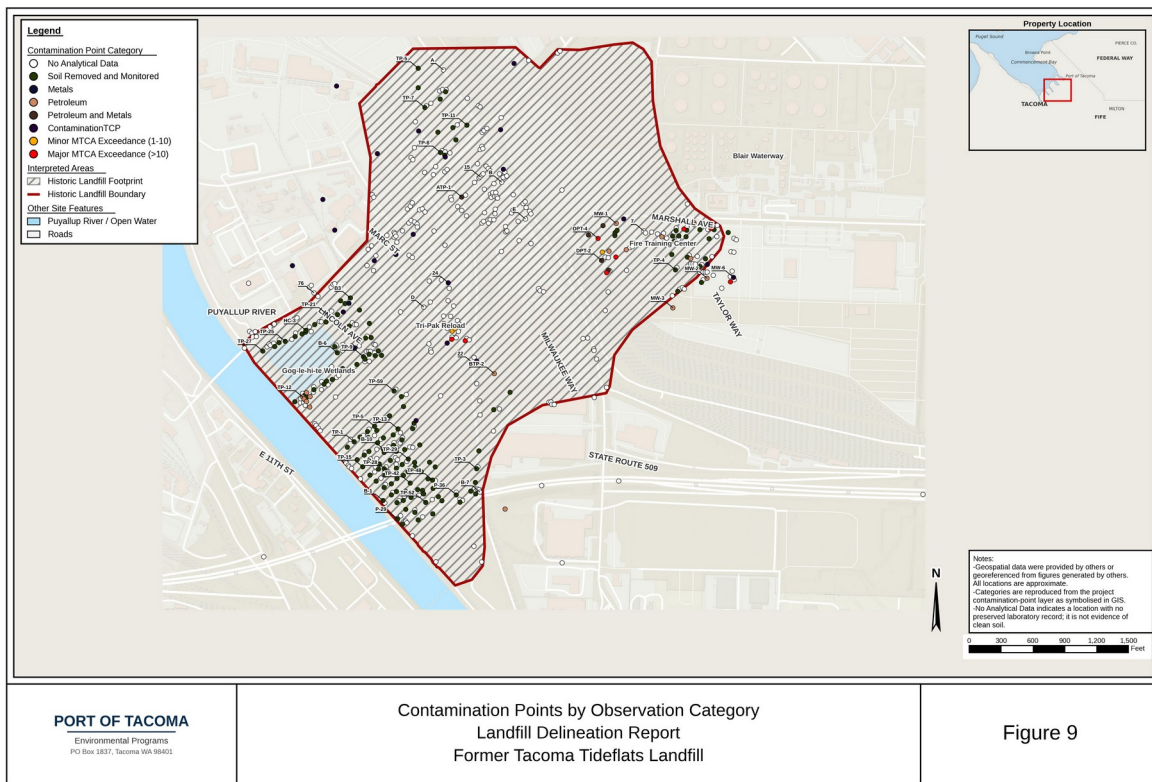


Figure 9. Contamination Points by Observation Category. 559 mapped investigation locations in the nine categories used in the project GIS.

Figure 10. Contaminant Concentration Ratios. Where a laboratory result can be expressed against a cleanup level, the ratio makes the difference between the two clusters plain. The Marshall Avenue reach carries the densest run of exceedances but at moderate ratios, mostly between one and seven times the applicable level. The single highest ratio on the site is not in that reach at all, but near Tri-Pak Reload. A second group of detections carries no ratio because the record identifies the contaminant but not a comparable concentration; these are drawn as hollow categorical symbols and should be treated as unquantified rather than as low. Symbol area, not radius, is proportional to the ratio, so the 225 result reads as large without dominating the sheet.

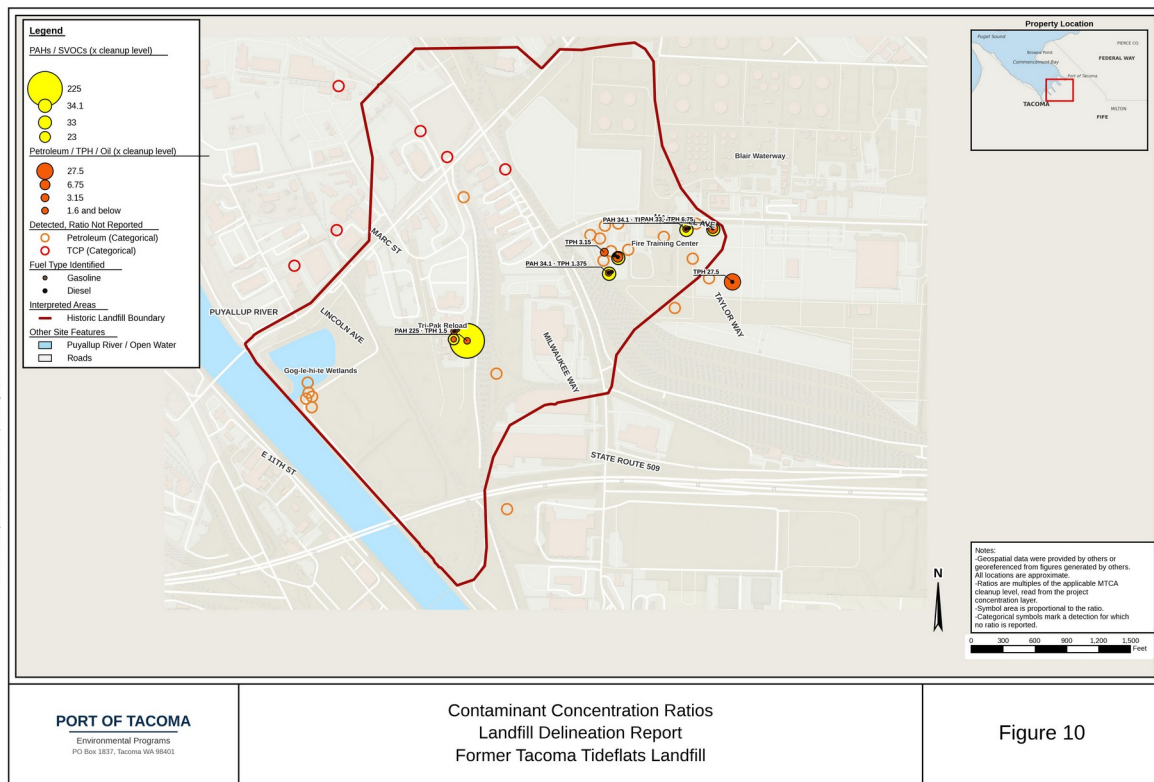


Figure 10. Contaminant Concentration Ratios. PAH/SVOC and petroleum results as multiples of the applicable MTCA cleanup level; symbol area tracks the ratio.

Figure 11. MTCA Soil Screening. Ecology cleanup site points within the study area were screened separately for soil, groundwater and sediment against Model Toxics Control Act cleanup levels. Soil carries the largest number of exceedances of the three media. A number of points remain unevaluated and are shown as status unknown; as elsewhere in this record, unknown means the screening has not been performed or the result was not preserved, not that the point is clean.

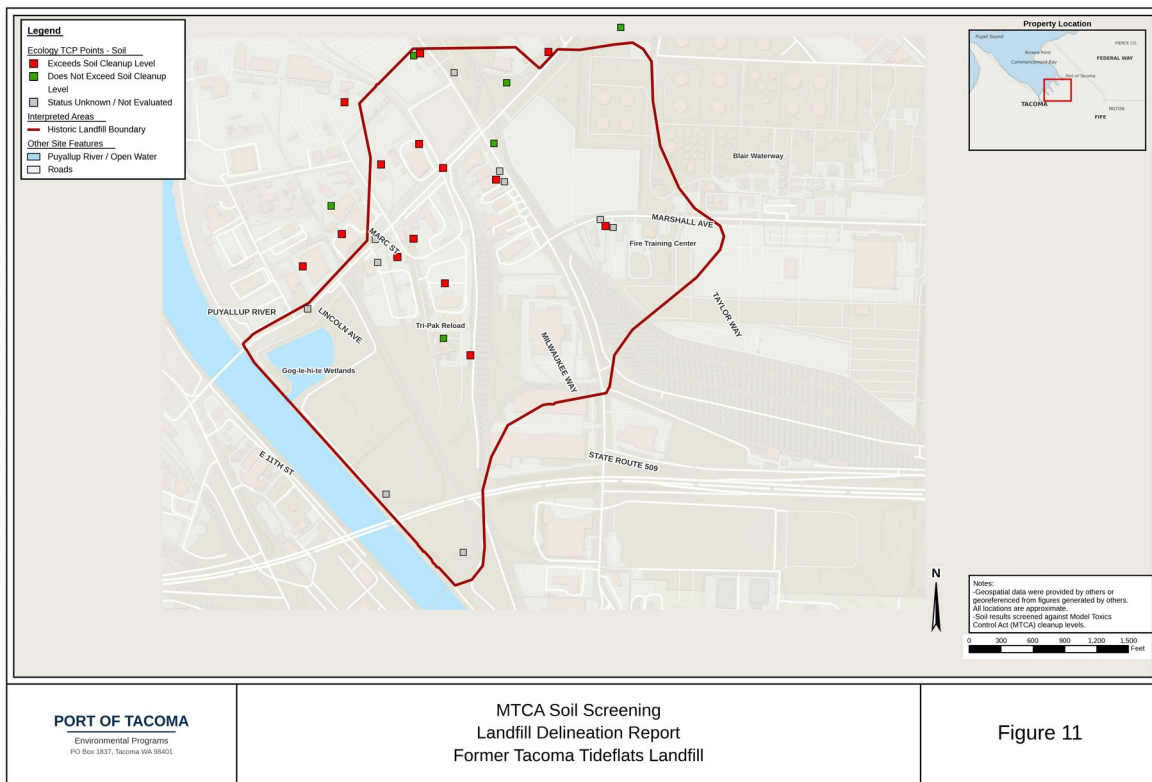


Figure 11. MTCA Soil Screening. Ecology cleanup site points screened against MTCA soil cleanup levels.

Figure 12. MTCA Groundwater Screening. Groundwater carries the second largest number of exceedances. Benzene is notable for a low soil exceedance rate but a comparatively high groundwater rate, which is consistent with the shallow groundwater conditions found across much of the site. Shallow groundwater is also a recurring constraint in the habitat restoration record, where it appears alongside refuse and fill as a condition to be designed around.

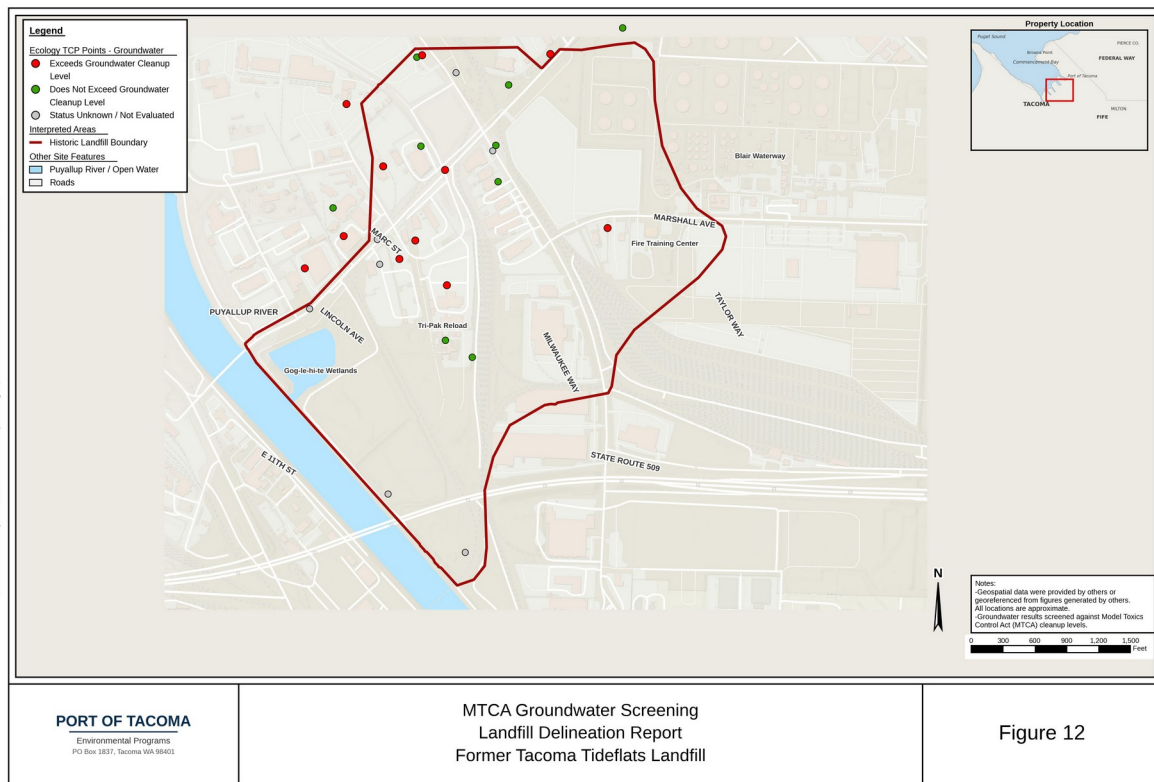


Figure 12. MTCA Groundwater Screening. Ecology cleanup site points screened against MTCA groundwater cleanup levels.

Figure 13. MTCA Sediment Screening. No exceedance was recorded among the sediment points screened. That result reflects the points that have been screened rather than the sediment condition of the site as a whole, and the screened set is small relative to the soil and groundwater sets.



Figure 13. MTCA Sediment Screening. Ecology cleanup site points screened against MTCA sediment cleanup levels.

Figure 14. Gog-le-hi-te Habitat Investigation Points. The Port developed the initial Gog-le-hi-te estuary in 1985 on former landfill ground near the Puyallup River, and expansion work followed in 2003 and again in 2007 and 2008, removing considerable volumes of soil, refuse, concrete and other debris. Investigations in the southwestern portion of the footprint document landfill debris, refuse layers, construction waste, wood debris, shallow groundwater and contamination associated with historical landfill materials. Habitat construction and the monitoring that followed are the reason so many locations in this part of the site are recorded as cleaned up or below cleanup levels. Planning continues for a future Gog-le-hi-te III area and for connections between the existing and future habitat areas.

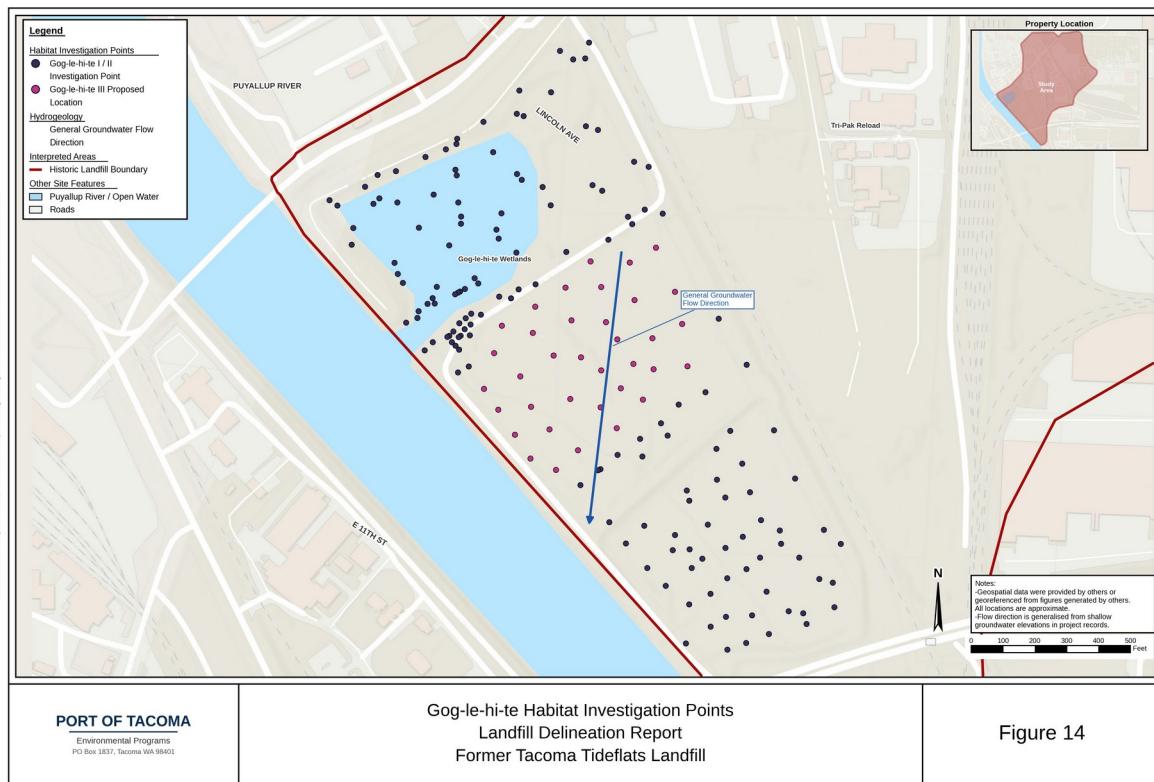


Figure 14. Gog-le-hi-te Habitat Investigation Points. Restoration area sampling coverage, proposed Gog-le-hi-te III locations, and generalised shallow groundwater flow.

Figure 15. Methane Monitoring Bar Holes. Methane generated by decomposing organic waste remains a long-term monitoring concern. Inspections between 1997 and 2004 documented high readings, including values at or above 100 percent of the lower explosive limit in some events, and a further elevated reading was recorded in 2014. Later records generally show lower concentrations, which has supported a proposal to move to annual methane monitoring while retaining quarterly site inspections. In 2023 low concentrations of landfill gas were detected at seven sampled bar-hole locations. Wetland expansion work altered monitoring access in places, so gaps in the bar-hole record are not always gaps in the condition being monitored. The compiled chronology extends through the May 2026 second-quarter inspection.

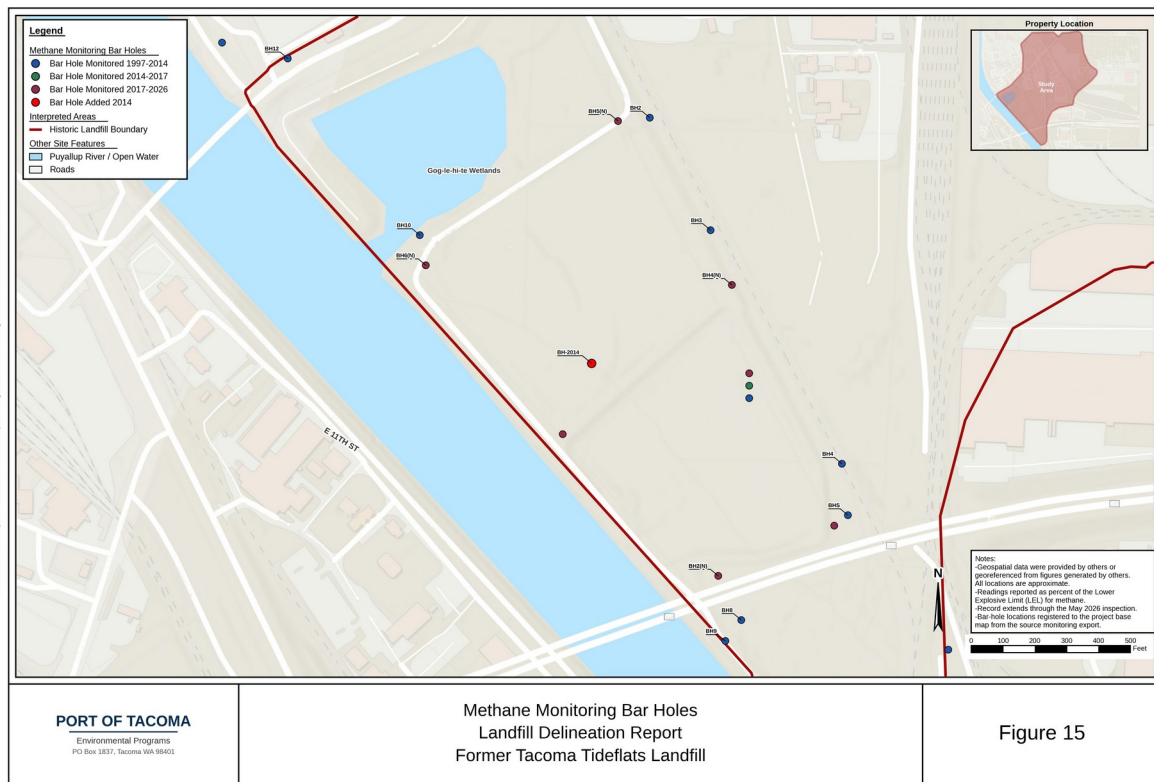


Figure 15. Methane Monitoring Bar Holes. Landfill gas monitoring network and inspection record, 1997 to 2026.

Location: The site occupies the Tacoma Tideflats between the Puyallup River and Taylor Way, north of State Route 509 and either side of Lincoln Avenue, in Tacoma, Washington. The interpreted footprint spans roughly 6,700 feet east to west and 5,400 feet north to south, covering more than 200 acres. It includes the Gog-le-hi-te Wetlands at its southwestern edge, the Tri-Pak Reload property at its centre, and the Fire Training Center and Marshall Avenue reach in the east. The area lies on the low-lying floodplain and tidal environment of the Puyallup River delta, and the Blair Waterway lies immediately to the northeast.

Timeline

- **Early 1940s:** The City of Tacoma begins municipal refuse disposal on the Tideflats, using waste both for disposal and to raise ground elevations for future industrial development.
- **1940s-1964:** Municipal solid waste continues to be deposited across the landfill area. Open burning is reported as a common waste-management practice of the era.
- **1964:** The Lincoln Avenue Landfill ceases operation. Soil and fill are placed over the refuse and industrial redevelopment expands across the footprint.
- **1970s-1980s:** Former landfill properties are redeveloped for rail yards, warehouses, trucking facilities, industrial storage yards, and transportation corridors.

- **1980s:** EPA and Ecology investigate the former landfill. EPA determines the site does not qualify for the National Priorities List.
- **1985:** Portions of the footprint near the Puyallup River are converted to estuarine habitat as the Gog-le-hi-te Wetlands project.
- **1988:** Port-controlled land near the East Blair Waterway is conveyed to the Puyallup Tribe through the Puyallup Land Claims Settlement.
- **1993:** The Tacoma-Pierce County Health Department publishes its original closed-landfill study, the first interpreted map of the boundary.
- **1997-2004:** Quarterly inspections document high methane readings at several bar holes, including values at or above 100 percent of the lower explosive limit.
- **2001:** A landfill characterisation study is completed for the Milwaukee Railyard area, evaluating waste composition, thickness, and extent.
- **2003:** The U.S. Army Corps of Engineers expands habitat restoration areas near Gog-le-hi-te and excavates portions of historical fill during construction.
- **2006:** Gog-le-hi-te III habitat project investigations document extensive landfill debris, refuse layers, construction waste, wood debris, shallow groundwater, and contamination associated with historical landfill materials.
- **2007-2008:** The Port acquires additional properties within the interpreted footprint, including Parcel 95, and continues wetland mitigation and restoration within former landfill areas. Wetland expansion work removes substantial soil, refuse, concrete, and other debris.
- **2010:** A detailed environmental investigation at 1620 Lincoln Avenue documents landfill debris, shallow groundwater, petroleum contamination, elevated metals, ash, sawdust, glass, brick, metal, plastic, and organic refuse extending approximately 9 to 10 feet below ground surface.
- **2010:** The Health Department's revised mapping differs from the 1993 interpretation and reflects continuing uncertainty about the true extent of disposal.
- **2017:** An environmental investigation at 1620 Lincoln Avenue encounters a surface layer of sand and gravel fill extending approximately 2.5 to 4 feet below ground surface, over landfill deposits containing ash, sawdust, organic matter, glass, wood, brick, and plastic.
- **2020-2025:** Additional investigations associated with the Taylor Way and Alexander Avenue Fill Area, Gog-le-hi-te restoration areas, and landfill-related parcels are carried out to evaluate soil, groundwater, and vapour, fill materials, and landfill conditions.
- **2023:** Low concentrations of landfill gas are detected at seven sampled bar-hole locations within portions of the former landfill footprint.
- **2024:** Port planning and development documents identify Parcel 37 as part of the former City of Tacoma landfill and incorporate potential refuse, contamination, and settlement considerations into future planning efforts.

- **2025:** Ecology records identify continued monitoring of landfill-related environmental conditions, including methane generation, groundwater conditions, and potential contaminant-migration pathways associated with buried refuse.
- **2026:** The compiled investigation-point inventory, depth-interval tables, contaminant taxonomy, cross sections, and historical chronology are delivered as an integrated GIS and documentation package. The historical record extends through the May 2026 second-quarter inspection.

Use: The landfill operated as a municipal refuse disposal site from the early 1940s until 1964, and waste placement also served to raise ground elevations for industrial development. Following closure the area was capped with soil and fill and redeveloped for industrial use. Today the footprint is held by the Port of Tacoma, private industrial operators, railroads, public agencies, and the Puyallup Tribe. Present uses include rail yards, warehousing, trucking and transportation facilities, industrial storage yards, roadways, and habitat restoration, and ten Port parcels intersect the interpreted footprint.

Contamination (or Prior Remediation): The compiled inventory holds 559 mapped investigation locations. Analytical coverage is uneven: a substantial share of the record consists of historical boreholes, monitoring wells, and test pits with no preserved laboratory results, which is a gap in the record rather than evidence of clean soil. Where results exist, the most common contaminants exceeding cleanup criteria are petroleum hydrocarbons, metals, polycyclic aromatic hydrocarbons, and polychlorinated biphenyls, with chlorinated solvents and PFAS at some locations. Confirmed exceedances concentrate in two areas: the Marshall Avenue and Fire Training Center reach in the east, and a cluster along the Lincoln Avenue corridor. Expressed as ratios, the eastern reach carries the densest run of exceedances but at moderate values, mostly one to seven times the applicable cleanup level, with petroleum and PAH results occurring together and gasoline or diesel identified at several points. The single highest ratio on the site sits apart from that cluster: a PAH result at 225 times the cleanup level near Tri-Pak Reload, paired with a petroleum result of only 1.5. Landfill debris, refuse layers, tar materials, and contaminated fill recur throughout the subsurface record and are commonly associated with the highest observed concentrations. Screening Ecology cleanup site points against Model Toxics Control Act cleanup levels shows soil carrying the largest number of exceedances, groundwater the second largest, and no exceedance among the sediment points screened.

Remediation (or Final Clean Up): The Lincoln Avenue Landfill is recorded by Ecology under CSID 3645 and FSID 1240 with a status of Awaiting Cleanup and a SHARP ranking of Medium. Thirteen further Ecology cleanup sites fall within or immediately adjacent to the footprint, a mixture of landfill-era conditions and later industrial releases, and several remain open; the practical consequence is that no single cleanup decision governs the former landfill, and obligations depend on which cleanup site record applies. Remediation to date has been substantial but localised. Wetland expansion work in 2003 and

again in 2007 and 2008 removed considerable volumes of soil, refuse, concrete, and other debris, and soil removal during habitat construction and the monitoring that followed account for the large number of southwestern locations now recorded as cleaned up or below cleanup levels. Landfill gas remains a long-term monitoring concern: inspections between 1997 and 2004 documented high readings including values at or above 100 percent of the lower explosive limit, a further elevated reading was recorded in 2014, and later records generally show lower concentrations, supporting a proposal to move to annual methane monitoring while retaining quarterly site inspections. Planning continues for a future Gog-le-hi-te III area and for connections between the Gog-le-hi-te I, II, and future habitat areas.

Cost Recovery (if applicable): No cost-recovery action is recorded for the former landfill in the compiled project record. The Taylor Way and Alexander Avenue Fill Area partly overlaps the eastern portion of the study area; that work proceeds on its own schedule, with a data-gap report issued in 2024 and a draft remedial investigation and feasibility study issued in 2025, and because the two footprints overlap, TWAAFA data are an important input to landfill delineation and any landfill-area conclusions should be checked against the current TWAAFA record.

Record and Methods: The delineation draws on a compiled investigation-point inventory built from parcel folders, consultant reports, Ecology cleanup-site data, Port maps and Tacoma-Pierce County Health Department records. Duplicate records were deliberately retained at the inventory stage for quality assurance. Because report identifiers repeat across parcels and reports, the working key is parcel plus report ID plus source report plus source object ID rather than report ID alone. Depth intervals were merged onto the point inventory on that composite key with match-confidence fields preserved, and the result confirms that the mapped inventory is considerably broader than the depth-resolved analytical record. The principal environment was ArcGIS Pro with Python and ArcPy, supported by Excel and CSV for schema cleanup and contaminant flags, Word and PDF for source documentation, and SharePoint and OneDrive for document management. Cross sections A-A' through E-E' were produced by an ArcPy workflow that measures station along each section line, builds the interval blocks, and places labels, leaders, grids and axes.

Data Quality and Recommended Controls: This compilation is a screening and delineation tool assembled from decades of records of varying quality, and the following controls should be carried forward by anyone extending or relying on it. Use composite keys rather than report ID alone. Preserve the original source object ID, document, page, figure and table references so every mapped point can be traced back. Keep locations, depth intervals, status, documents, inspections and gas readings in separate related tables, and keep field observations separate from analytical detections and exceedances. Preserve filename-date conflicts and optical character recognition uncertainty as explicit quality-assurance fields rather than silently resolving them. Treat points outside the working boundary as review evidence, not as automatic grounds for expanding it. Label vertical exaggeration on every cross

section, and distinguish schematic from measured sections. All locations shown in this fact sheet are approximate: geospatial data were provided by others or georeferenced from figures generated by others.

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Source documents are held in the Port of Tacoma project data folders and the project SharePoint library. The complete source-document register, hyperlink index, and AcrPy script archive are maintained separately in the project record and are not reproduced here.